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Caesarean Sections in Teaching Hospitals: Systematic Review and Meta-analysis

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Caesarean Sections in Teaching Hospitals: Systematic Review and Meta-analysis

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Keywords

caesarean section, teaching hospitals, access to care, medical practice variation, health services, quality of care, overuse

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Abstract

Objective: The aim of this study is to determine the odds of caesarean section in all births in teaching hospitals as compared with non-teaching hospitals.

Setting: Teaching and non-teaching hospitals in 22 countries.

Participants: Women at birth.

Intervention: Caesarean section.

Primary and Secondary Outcome Measures: The primary outcome measure is the adjusted odds ratio of caesarean section in teaching and non-teaching hospitals. The secondary outcome is the crude odds ratio of caesarean section in teaching and non-teaching hospitals.

Results: Based on 18 studies, the adjusted odds for caesarean section in teaching hospitals as compared to non-teaching hospitals were 0.90 (95%CI 0.58 to 1.40), with high heterogeneity between studies ($\tau 2=0.866$). The crude odds ratio, calculated using 27 studies, was 1.08 (95%CI 0.76, 1.54) again with high heterogeneity between studies ($\tau 2=0.866$).

Conclusions: Overall, there is no significant difference in the likelihood of caesarean sections between teaching and non-teaching hospitals. Nevertheless, teaching hospitals, in several subgroups, have a lower or higher likelihood of caesarean sections. Therefore, we recommend a more precise examination of forces sustaining these trends.

Trial Registration: Prospero - CRD42020158437

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Strengths and limitations of this study

- The major strengths of our systematic review and meta-analysis include a broad literature search, data extraction performed by more than one reviewer, an exploration of study characteristics as a potential source of variation between studies, and quality assessment using QUIPS.
- Some additional strengths of this study are the very large sample size and the inclusion of many hospitals over broad geographic regions.
- Potential limitations relate to differences in the characteristics of compared hospitals, study populations, type of data used, types of CS analysed, and variables used for adjustment in statistical analyses across studies.
- We also lack information on hospital resources, infrastructure and staffing of compared hospitals which could have been useful for a more detailed analysis of the underlying factors that may contribute to differences among teaching and non-teaching hospitals.

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Introduction

Caesarean sections (CS) are a key surgical procedure in improving long-term health outcomes in high-risk births.[1 2] Key to providing high quality healthcare is also avoiding the risks associated with using this procedure in low-risk pregnancies. The WHO in 1985 suggested an optimal rate of 10-15%,[3] however the rate has continued to increase, rising to an estimated 21.1% worldwide in 2015.[4] Furthermore, there is a large degree of variation observed, for example, the rate in different countries ranges from dangerously low, i.e. below 10%, to extraordinary high rates, e.g. 58.1% in the Dominican Republic.[4] Within country variations have also been observed in many countries including the United States.[5] More recently, in Italy, the 2018 data concerning Nulliparous, Term, Singleton, Vertex CS shows variation between 8.4% and 44.4% among health authorities, with significant and constant intra-regional variation.[6]

It is highly unlikely that such dramatic differences are explained solely by clinical factors. This may stem from a variety of factors, such as insurance status or socioeconomic background.[7-11] Factors affecting patient or clinician preferences may also be responsible for variations in healthcare delivery. Patients prefer caesareans for cultural or experience related reasons. Physicians can be less skilled in relation to natural birth, in comparison to, for example, midwives.[12-19] Hospital factors or insurance type may also have an effect.[8 20 21]

Another relevant factor is hospital teaching status. In general, teaching hospitals are known to have higher overall rates of surgery, which is perhaps reflective of more complex caseloads.[22-24] For this reason, it may be intuitively assumed that teaching hospitals have a higher proportion of high-risk births and, as such, higher likelihood of CS as well. Teaching hospitals are generally associated with a higher standard of care and better outcomes,[24-30] although there are also studies which reported that for certain outcomes there is no significant difference compared with general hospitals.[31] It may be argued that higher quality care can also translate to a lower likelihood of caesarean delivery as a result of fewer unnecessary procedures. It has been suggested are that teaching hospitals generally have access to a certain level of technology, and that it is easier to apply protective measures, such as requiring a second opinion on CS procedures.[25] The goal of this review is to carry out meta-analysis on available studies

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which give the odds of caesarean section in teaching and non-teaching hospitals. Thereby we aim to determine whether there is a significant difference, evaluate the effect size and variations among different subgroups and thereby understand the differences of delivery care provision among teaching and non-teaching hospitals.

Materials and Methods

This meta-analysis was constructed as per previous meta-analyses carried out by members of our group[20 32 33] and in accordance with the structure suggested by Dekkers and co-workers,[34] the Preferred Reporting Items for Systematic reviews (PRISMA) guidelines,[35] and Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines for meta-analysis of observational studies.[36] We produced a protocol, which was submitted and registered with Prospero (Prospero ID: CRD42020158437).

Eligibility Criteria, Information Sources, Search Strategy

In order to get the widest spread of data possible we searched multiple databases - CINAHL, Scopus, Web of Science, The Cochrane Library and PubMed from the beginning of records until May 2020. The search terms were based on a four-level strategy, with terms related to caesarean sections, terms related to other variables, in this case hospital teaching status, terms related to area of research i.e. health system variation etc., and terms related to study design.

Study Selection

The primary criterion for study inclusion was the presence of an odds ratio (OR) for caesarean sections in teaching and non-teaching hospitals. Reports included other data (i.e. regression coefficients, case data by type of hospital and delivery outcome) allowing the ORs to be calculated were also included. Study inclusion was also contingent on study design, targeting in this case reviews, systematic reviews, cohort studies and cross-sectional studies. Studies were excluded to remove overlapping data. The eligibility of each study was assessed independently by at least two researchers, and disagreements were resolved by consultation.

Data Extraction

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The data extraction was carried out by at least two independent researchers for each study. Another reviewer performed a final review of all extracted data. Disagreements were resolved by discussion. A data extraction spreadsheet was constructed to facilitate the process, and included a number of pertinent data, such as publication data, OR, confidence limits, main determinant being studied and cohort size, among other data. For purpose of classification of population of each study we extracted exclusion/inclusion criteria data that were used by authors in order to specify their study population. Such data enabled us to classify population for each study based on Robson system. Robson's ten groups classification system is based on five main population features including obstetric history, onset of labour, presentation of features, number of neonates, and gestational age.[37] This is a standardized comparison method that is helpful and widely used to analyse trends and determinants of CS.[38]

Assessment of Risk of Bias

The risk of bias for included studies was assessed in terms of six domains – study participation, study attrition, prognostic factor measurement, study confounding, and statistical analysis and reporting, using the Quality in Prognosis Studies (QUIPS) tool.[39] Although it is a standardised tool for prognostic studies it fits very well with overall structure of included papers and it has already been used it in two methodologically similar reviews.[32,33]

Data Synthesis

The data was subjected to statistical analysis to give an overall OR across all included studies. The ORs were combined using standard inverse-variance random-effects meta-analysis, where an OR lower than one shows a lower likelihood of CS in teaching hospitals than non-teaching hospitals.[40] The teaching hospitals were considered hospitals with academic centre/teaching status, i.e. hospitals where residency training and/or academic research takes place. The non-teaching hospitals where other hospitals such as general hospitals or community hospitals with no such (academic/teaching) status.

Subgroup analysis by country, study design, period of data collection, parity, inclusion of women with previous CS, population by Robson groups, type of CS analysed, type of data used and level of (QUIPS)

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risk of bias domains was used to examine between-study heterogeneity and χ^2 test used to calculate p values for interaction among subgroups.

Statistical analysis was carried out using STATA, release V.15 (StataCorp).

Patient Involvement

No patients were involved in this study. We used data from published papers only.

Results

Study Selection

We identified a total of 4582 records: 1522 from PubMed; 534 from Scopus; 1049 from the Cochrane Library, 1133 from Web of Science; 344 from CINAHL and 68 from manual search (Figure 1). We removed 698 duplicates. 3952 records were screened for eligibility. We performed full-text examination on 908 records. We excluded 827 that did not report hospital teaching status, 46 that were otherwise irrelevant and three records with overlapping populations. Finally, 32 records describing 32 separate studies,[16 25 41-70] including 18.5 million births in 22 different countries, were included in the review and meta-analysis. Of these, 18 studies were included in the meta-analyses of adjusted odds ratios and 27 studies were used in the meta-analyses of crude odds ratios.

Study Characteristics

Characteristics of included studies are shown in Table 1 and Web Appendix. Most included studies were retrospective studies, with some prospective studies also included. The relevant studies were from a variety of countries and health systems. Sample size varied from 214 to 8.48 million births. The oldest study was from 1991, but the vast majority of included studies were carried out post-2000. Case exclusion criteria and covariates for statistical adjustment also varied considerably.

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Author	Year	Country/region	Study design	Number of cases	Number of hospital units	Year of data collection	Population by Robson groups	Sampling	Type of CS analysed
Oleske et al.	1991	United States	Retrospective	130,249	198	1986	2 and 4	Consecutive	Planned
Sanchez-Ramos	1994	United States	Prospective	945,707	236	1992	All	Random	Any
Braveman et al.	1995	United States	Retrospective	217,461	Not reported	1990-1991	1 to 2 and 6	Consecutive	Any
Irwin et al.	1996	United States	Retrospective	3,603	75	1987-1989	1 to 2 and 6	Consecutive	Any
Chanrachakul et a	2000	Thailand	Retrospective	1,073,403	227	1999	All	Random	Any
Murray	2000	Chile	Prospective	504	3	1995-1997	All	Random	Any
David et al.	2001	France	Retrospective	40,512	149	1994-1998	1 to 2 and 6	Consecutive	Any
Garcia et al.	2001	United States	Retrospective	63,143	36	1996	All	Consecutive	Any
Korst et al.	2005	United States	Retrospective	443,532	288	1995	1 and 2	Consecutive	Any
Linton et al.	2005	United States	Retrospective	53,215	66	2002	All	Consecutive	Any
Carayol et al.	2007	France	Prospective	6,091	174	2001-2002	2 and 4	Random	Planned
Tamim	2007	Lebanon	Retrospective	18,837	9	2001-2002	1 to 2 and 6	Consecutive	Any
Xirasagar et al.	2007	Taiwan	Retrospective	739,531	942	1997-2000	2 and 4	Consecutive	Planned
Chen et al.	2008	Taiwan	Retrospective	200,207	Not reported	2004	All except 8	Consecutive	Any
Liu et al.	2008	Taiwan	Retrospective	247,751	Not reported	1998-2002	All except 8	Consecutive	Any
Taljaard et al.	2009	Latin America	Prospective	95,164	119	2004-2005	1 to 5	Random	Any
Zhang et at.	2010	United States	Retrospective	206,969	19	2002-2008	1 to 4 and 6	Consecutive	Any
Snyder et al.	2011	United States	Retrospective	244,464	129	2006-2007	1 to 5	Consecutive	Any
Aman et al.	2014	Ethiopia	Retrospective	17,761	6	2011	All	Consecutive	Any
Stivanello et al.	2014	Italy	Retrospective	213,539	36	2005-2010	All	Consecutive	Any
Bommarito et al.	2016	United States	Retrospective	8,485,034	1,051	1998-2010	1 to 4	Random	Emergency
Kim et al.	2016	Korea	Retrospective	1,289,989	674	2011-2014	All	Consecutive	Any
Meda et al.	2016	Burkina Faso	Retrospective	668	66	2009-2010	All	Random	Any
Oner et al.	2016	Tukey	Prospective	214	4	2012	1 to 2 and 6	Random	Any
Faisal-Cury et al.	2017	Brazil	Prospective	757	10	2005-2006	All except 8	Random	Any
Lorthe et al.	2017	France	Prospective	560	546	2011	6 and 7	Consecutive	Any
Clapp et al.	2018	United States	Retrospective	1,657,495	1,206	2013	1 to 4	Consecutive	Any
Zbiri et al.	2018	France	Retrospective	291,449	11	2008-2014	All	Consecutive	Any
Zimmo et al. A	2018	Palestine	Prospective	18,908	3	2016-2017	All	Consecutive	Any
Zimmo et al. B	2018	Palestine	Prospective	9,567	2	2015-2016	1 to 4	Consecutive	Emergency
del Carmen et al.	2019	United States	Retrospective	1,855,675	Not reported	2006-2010	1 and 3	Consecutive	Emergency
Hoxha et al.	2019	Kosovo	Prospective	859	5	2015	1 and 2	Consecutive	Any

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Risk of Bias of Included Studies

The main potential source of bias is study confounding, where 17 of the 32 studies included had a high risk of bias and 9 studies had a moderate risk of bias. With respect to study participation, 4 studies had a moderate risk of bias, with the remainder having a low risk of bias. A low risk of bias in all studies was reported for study attrition, prognostic factor measurement and outcome measurement, while only two studies had a moderate risk of bias for the statistical analysis and reporting domain. Thus, the main source of potential bias was lack of control for known confounders in included studies.

Synthesis of Results

Figure 2 presents meta-analyses for primary outcome measure, i.e. adjusted ORs of teaching hospitals in comparison with non-teaching hospitals. CS was 0.90 times as likely in teaching hospitals (95% CI 0.58 to 1.40) with high heterogeneity between studies ($\tau 2=0.866$), although the difference was not significant.

Subgroup analyses of adjusted odds ratios in teaching and non-teaching hospitals (Figure 3) showed large variations. Only two of the countries where studies were available had significantly higher likelihood of CS in teaching hospitals, where Korea (OR=4.95, 95%CI 4.82 to 5.09) showed a large difference, with a smaller difference in Lebanon (OR=1.40, 95%CI 1.30 to 1.50). Conversely, in Kosovo (OR=0.46, 95%CI 0.24,0.89), Palestine (OR=0.53, 95%CI 0.37,0.77), Taiwan (OR=0.86, 95%CI 0.82,0.91) and the United States (OR=0.87, 95%CI 0.82,0.91), the likelihood of CS in teaching hospitals was lower. Significant difference was also observed by study design, particularly in prospective studies, which found that CS was 0.62 times as likely in teaching hospitals (95%CI 0.45, 0.86). The likelihood of CS in teaching hospitals was significantly lower in studies carried out before 2000 (0.83, 95%CI 0.74,0.93).

Indication for CS and pregnancy types led to variation in CS rates between teaching and non-teaching hospitals. Teaching hospitals were significantly less likely to perform emergency CS (OR=0.66, 95%CI 0.46, 1.00). Mothers without previous CS were also significantly less likely to undergo CS in teaching hospitals (OR=0.79, 95%CI 0.72, 0.87). In subgroup analysis of population by Robson we see lower odds of CS in teaching hospitals which are significant for Robson groups 1 and 2 (OR = 0.65, 95%CI 0.45, 0.94), 1 and 3 (OR = 0.80, 95%CI 0.69,0.93), 6 and 7 (OR = 0.32, 95%CI 0.16, 0.66), and studies using

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all Robson groups except 8 (OR = 0.86, 95%CI 0.82, 0.91), with exception of studies that included all Robson groups in their analysis.

Figure 4 gives the secondary outcome, the crude ORs of CS for teaching and non-teaching hospitals. The crude odds were 1.08 in teaching, as compared to non-teaching, hospitals (95%CI 0.76 to 1.54), with high heterogeneity between studies ($\tau 2=0.866$).

Discussion

There was no overall significant difference between teaching and non-teaching hospitals in adjusted and subgroup analyses. The overall adjusted odds of receiving a CS were 0.90 (95% CI 0.58 to 1.40) for women admitted to a hospital with teaching status compared with women admitted to non-teaching hospitals. Subgroup analyses showed varied estimates in subgroups. Teaching hospitals carry out fewer CS in several countries, studies with prospective design, studies looking at data before 2000, for several study populations, emergency CS and women with no previous CS. In two countries, teaching hospitals perform more CS. The odds were 1.08 for crude estimates (95%CI 0.76 to 1.54).

The major strengths of our systematic review and meta-analysis include a broad literature search, data extraction performed by more than one reviewer, an exploration of study characteristics as a potential source of variation between studies, and quality assessment using QUIPS. Some additional strengths of this study are the very large sample size and the inclusion of many hospitals over broad geographic regions. Potential limitations which may have affected inconsistencies in the results, relate to differences in the characteristics of compared hospitals, study populations, type of data used, types of CS analysed, and variables used for adjustment in statistical analyses across studies. Moreover, we lack information on hospital resources, infrastructure and staffing, all of which could contribute to maternal outcomes and risk for CS.[5]

The findings in this study demonstrate the complexity of the interactions of many factors. [5] [16 71-74] The role of health system factors in general, and hospital factors in particular should not be ignored. Recently, meta-analyses examining the effect of hospitals and financial factors[20 32 33] have found that CS was 1.41 times as likely in for-profit hospitals as compared with non-profit hospitals (95% CI 1.24 to

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1.60),[20] 1.13 for privately insured women compared with women covered with public insurance (95% CI 1.07 to 1.18) [33] and 0.70 in uninsured as compared to privately and publicly insured women (95%CI 0.69, 0.72).[32]

The findings in our subgroup analyses indicate that the effort of hospital teaching status varies between countries and depends on study design and confounders considered in data analysis and year of data collection. Specific populations of mothers, and particular type of CS are also important variables in examining the effect of hospital variation. Patient populations in teaching versus non-teaching hospitals are generally very different in terms of underlying medical conditions and obstetric risk, which would likely drive any differences observed in odds for CS. Hence, intuitively, we would expect that odds for CS are likely to be higher in teaching hospitals, where more complex cases are often taken. Our study does not show this, and in several subgroups the opposite is observed. Variation in odds of CS in teaching hospitals for particular Robson groups hints at differences in implementation of clinical standards. Lower odds for emergency or women with no previous CS hints in a similar direction. The odds were higher in teaching hospitals in studies with a high risk of bias for confounding (Online Appendix), and lower in low or moderate risk of bias studies so it is likely true effects may well go in the direction of lower odds of CS for teaching hospitals, as seen in several subgroups. It is notable that before 2000 teaching hospitals performed significantly fewer CS than non-teaching hospitals. Studies have suggested that the lower odds in teaching hospitals may be due to availability of resources,[41 75] access to technology,[76] closer supervision and care,[57] stronger accountability[77], adherence to clinical standards,[16 25 54 78 79] and professional consultation opportunities.[25]

Our study contributes to understanding of the role of hospital related causes on the rates of CS. It may also provide an impetus for further research, particularly setting-specific studies, into underlying factors associated with teaching status, such as resource availability or adherence to clinical guidelines. In order to improve the quality of healthcare provision for deliveries, the interplay of factors associated with teaching status of hospitals must be examined, as well as outcomes, related to caesarean section rates but also other outcomes such as proportion of morbidity, mortality, poor-perinatal outcomes and particular forms of delivery (i.e. emergency and primary caesarean, vaginal birth after caesarean). Using

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classification by Robson groups or primary CS will help in understanding better the implications of results in future studies. Using variables with information such as infrastructure and other facets of hospital variation could be useful to elucidate more clearly the mechanism. Another way to understand the underlying factors related to hospital characteristics, which is being applied in Tuscany (Italy), is to put in place systematic and ongoing collection of PREMs (Patient-Reported Experience Measures) and PROMs (Patient-Reported Outcome Measures) [80] which could ensure continuous source of data to measure hospital performance outcomes.

Conclusion

We found no difference in the odds of receiving a CS for women admitted to a hospital with teaching status compared with women admitted in non-teaching hospitals. Nevertheless, teaching hospitals, in many subgroups, have a lower likelihood of caesarean sections, i.e. in most countries, low risk populations by Robson groups, women with no previous CS, studies with prospective design or moderate risk of bias in controlling for covariates. Lower or higher odds for CS among teaching hospitals hint at the effect of important aspects of healthcare delivery. Hence, further research would help to understand the interplay of underlying factors in individual settings which may prove useful for policy or medical practice efforts to ensure appropriate and efficient delivery care.

Author contributions

IH conceived and designed the study. BK, JB, FS performed the screening of studies. IH, EZ, KG, BK, JB, FS performed the data extraction and preparation. IH and EZ performed quality assessment of studies. IH has analysed the data. DAL, MB and GL have contributed to the interpretation of findings. IH, EZ and RA drafted the paper, which was critically reviewed and approved by all authors.

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All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: no support from any organization for the submitted work; no financial relationships with any organizations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

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Figure legends

Figure 1. The flow diagram of review

Figure 2. Adjusted odds ratios of caesarean section

Figure 3. Subgroup Analysis

Figure 4. Crude odds ratios of caesarean section

Supporting information

Appendix with detailed study information

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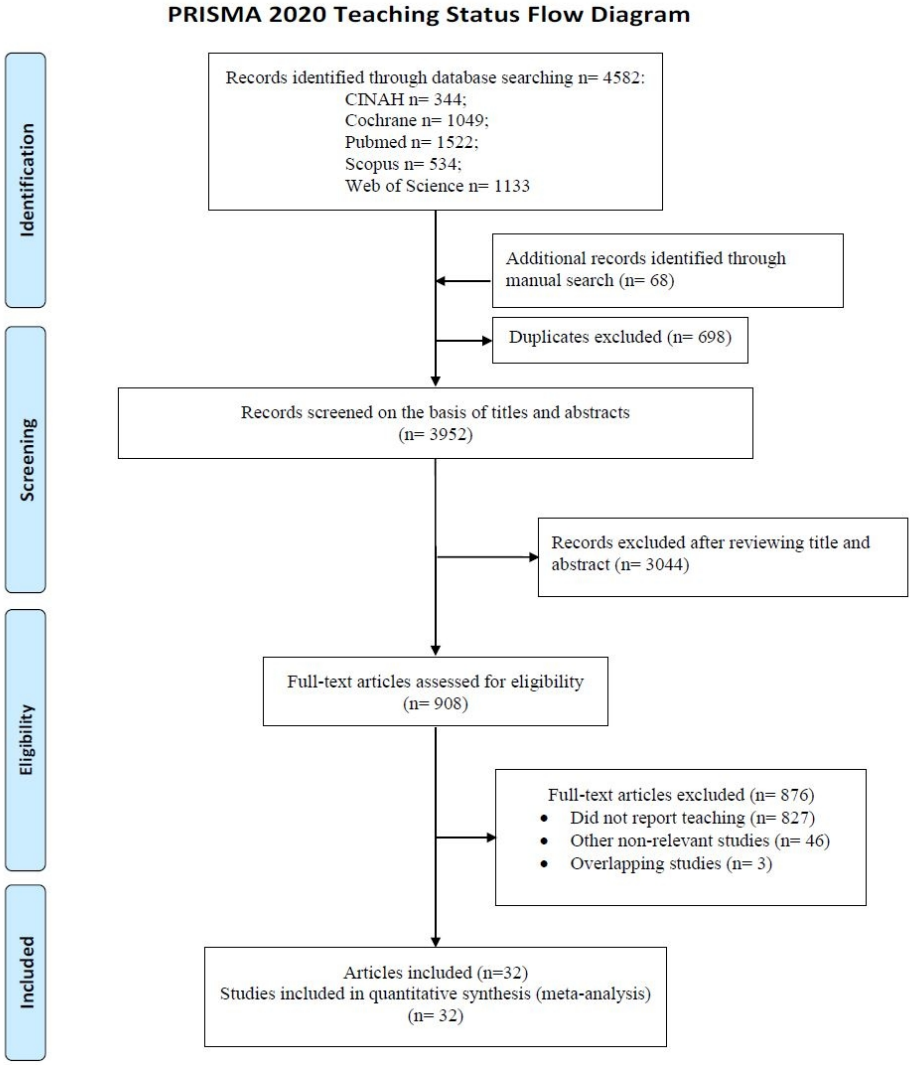


Figure 1. The flow diagram of review

177x190mm (144 x 144 DPI)

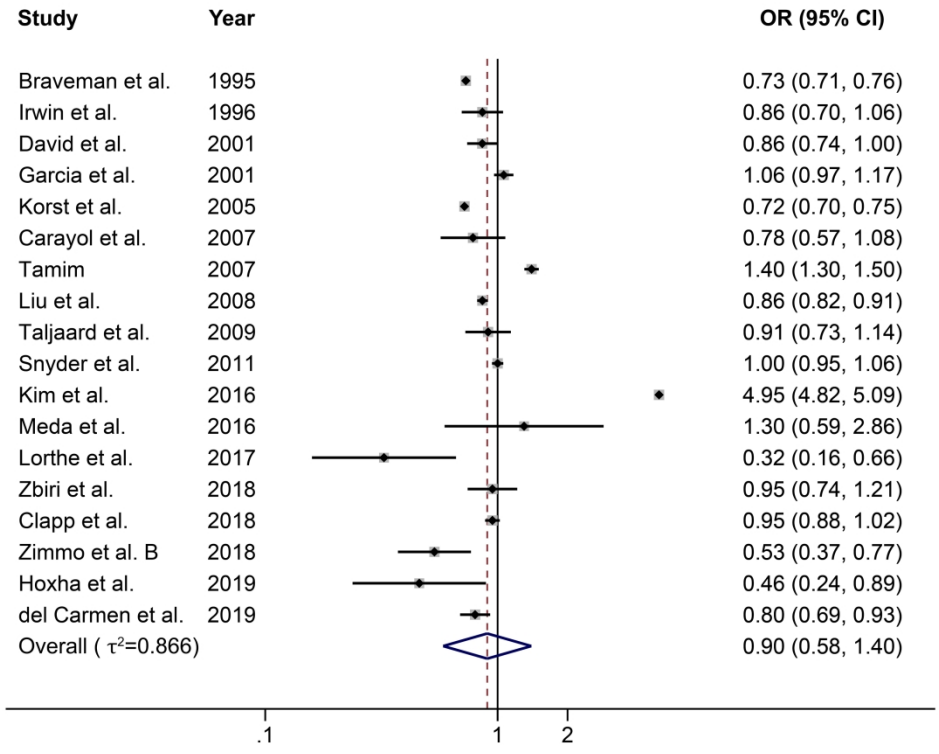


Figure 2. Adjusted odds ratios of caesarean section

282x247mm (300 x 300 DPI)

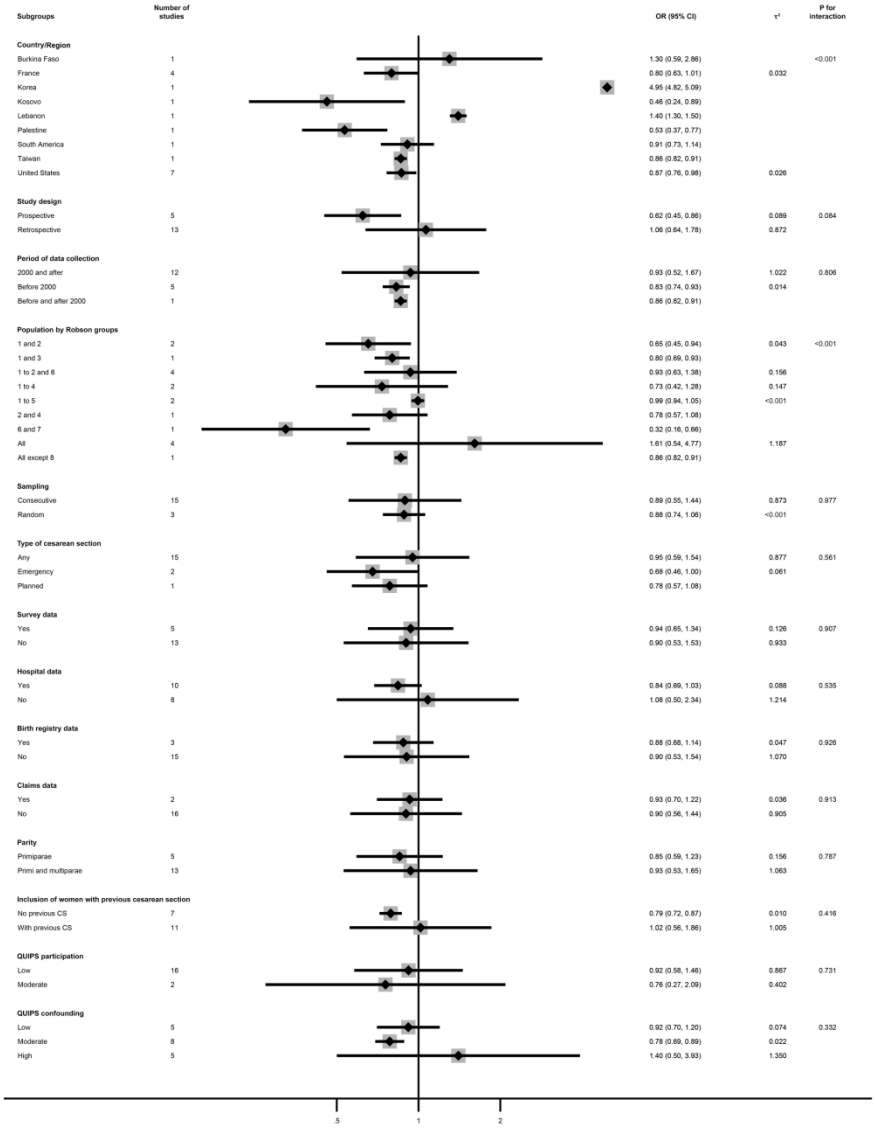


Figure 3. Subgroup Analysis

239x304mm (300 x 300 DPI)

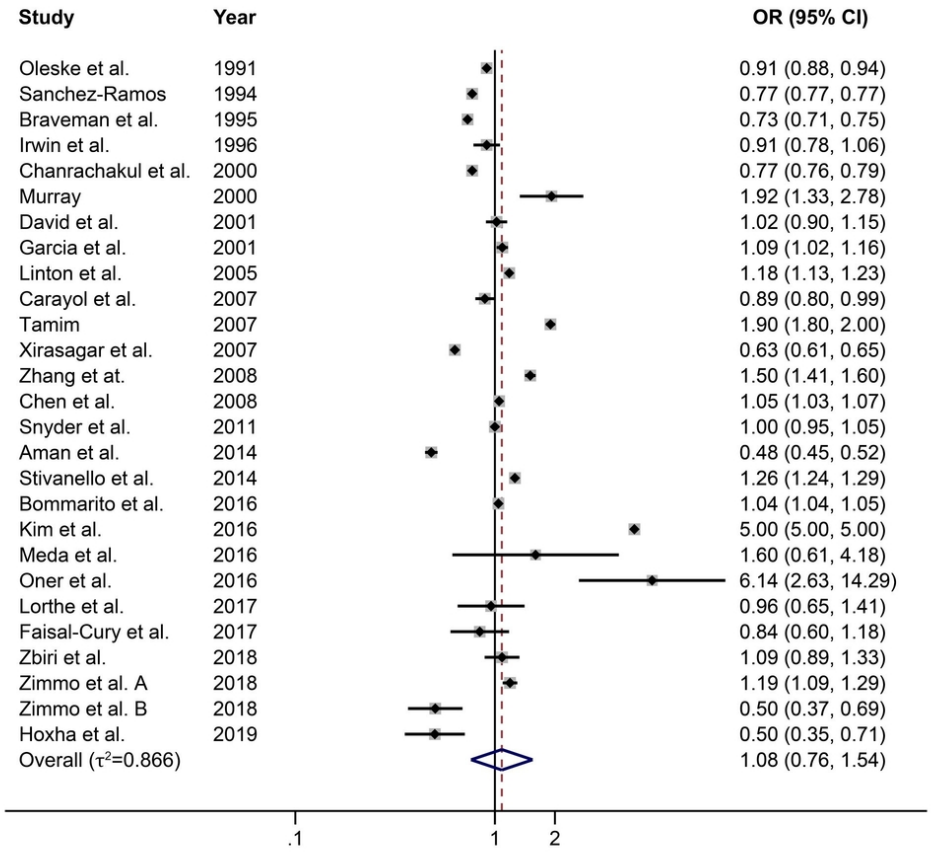


Figure 4. Crude odds ratios of caesarean section

85x83mm (300 x 300 DPI)

		Type of data used						
Author	Year	Survey	Hospital records	Birth certificates/registry	Claims data	Census data	Other	Source population
Oleske et al.	1991		x					All births in 198 hospitals in Illinois, United States
Sanchez-Ramos	1994	x						All births is 236 hospitals in United States
Braveman et al.	1995			x		x		All births in California, United States
Irwin et al.	1996		x					All births in 75 military hospitals in United States
Chanrachakul et a	2000	x						All births in 227 hospitals inThailand
Murray	2000	x						All births in 2 hopitals in Santiago, Chile
David et al.	2001		x			x		All births in 149 hospitals in France
Garcia et al.	2001		x		x			All births in 36 hospitals in Maryland, United States
Korst et al.	2005		x					All births in California, United States
Linton et al.	2005		x			x		All births in 66 military hospitals in United States
Carayol et al.	2007	x	x					All births in 138 hospitals in France and 36 hospitals in Belgium
Tamim	2007	x	x					All births in 9 hospitals in Beirut, Lebanon
Xirasagar et al.	2007				x			All births in 942 hospitals in Taiwan
Chen et al.	2008				x			All births in Taiwan
Liu et al.	2008					x		All births in Taiwan
Taljaard et al.	2009	x						All births in 119 hospitals in South America
Zhang et at.	2010		x					All births in 12 clinical centres in United States
Snyder et al.	2011			x				All births in 129 hospitals in Ohio, United States
Aman et al.	2014		x					All births in 6 hospitals in Addis Ababa, Ethiopia
Stivanello et al.	2014		x	x				All births in 36 hospitals in Emilia-Romagna, Italy
Bommarito et al.	2016		x					All births in 1051 hospitals in United States
Kim et al.	2016					x		All births in 674 hospitals in Korea
Meda et al.	2016	x						All births in Burkina Faso
Oner et al.	2016	x						All births in 82 hospitals in Burdur Province, Turkey
Faisal-Cury et al.	2017	x	x					All births in 10 hospitals in Sao Paolo, Brazil
Lorthe et al.	2017		x					All births in 546 hospitals in France
Clapp et al.	2018		x			x		All births in 1206 hospitals in 21 states in United States
Zbiri et al.	2018			x				All births in 11 hospitals in France
Zimmo et al. A	2018						x	All births in 3 hospitals in Gaza, Palestine
Zimmo et al. B	2018						x	All births in 2 hospitals in Palestine
del Carmen et al.	2019		x		x			All births in California, United States
Hoxha et al.	2019	x	x					All births in 5 public hospitals in Kosovo

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Covariates used for statistical adjustment																					
Maternal preconception status										Maternal clinical status				Foetus characteristics							
Ethnicity/Race	Educational level	Marital status	Economic status	Insurances status	Urban status	Weight	Height	Body mass index	Age	Parity	Previous caesarean section	Pre-existing (before pregnancy) conditi	Conditions developed during pregna	Gestational age	Birth weight	Other characteristics	Prenatal care	Birth characteristics	Provider characteristics	Other variables	Total number of covariates
																					0
				x	x				x		x	xx	xx			xx	x	xx	xx	xx	0
x	x	x	x	x					x			x			x	xx	x		xx	x	15
	x								x	x										x	4
x				x	x				x			xx	xx			x		x	xx	xx	17
																					0
			x		x				x		x								xx	xx	9
			x	x					x			xx	xx						xx		14
									x	x	x	x	x	x				xx	xx	xx	16
x	x	x	x					x	x	x			x						x	xx	13
x		x		x	x				x		x								xx	xx	11
	x		x	x	x				x						x		x		xx	xx	12
x	x	x											x						x	x	6
																					6
x				x					x										xx		7
											x	xx	xx			x				xx	12
									x										xx		5
		x							x	x				x					xx	xx	15
					x				x	x									xx	xx	7
																					0
				x					x										x	xx	5
																				xx	3
																					0
											x	xx	xx			x					6
																					0
										x	x	x	x			xx			xx	xx	10
				x				x	x	x			x	x	x		x			xx	11
																					0
									x	x	x			x	x	x		x	xx	x	17
																					0
																					0
	x				x			x	x		x	x					x	x		xx	10

+ One covariate adjusted for ++ Two or more covariates adjusted for

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QUIPS risk of bias					
Study Participation	**Study Attrition**	**Prognostic Factor Measurement**	**Outcome Measurement**	**Study Confounding**	**Statistical Analysis and Reporting**
low	low	low	low	high	low
low	low	low	low	high	low
low	low	low	low	moderate	low
low	low	low	low	low	low
low	low	low	low	moderate	low
low	low	low	low	high	moderate
low	low	low	low	high	low
low	low	low	low	low	low
low	low	low	low	low	low
moderate	low	low	low	high	low
low	low	low	low	high	low
moderate	low	low	low	low	low
low	low	low	low	moderate	low
low	low	low	low	high	low
low	low	low	low	moderate	low
low	low	low	low	low	low
low	low	low	low	high	low
low	low	low	low	moderate	low
moderate	low	low	low	high	low
moderate	low	low	low	high	low
low	low	low	low	moderate	low
low	low	low	low	high	low
low	low	low	low	high	moderate
low	low	low	low	moderate	low
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low	low	low	low	high	low
low	low	low	low	low	low
low	low	low	low	high	low
low	low	low	low	moderate	low
low	low	low	low	high	low
low	low	low	low	high	low
low	low	low	low	moderate	low

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4,5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6,7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7,8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8,9
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	(Appendix)
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8,9
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	9-11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11-12

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Caesarean Sections in Teaching Hospitals: Systematic Review and Meta-analysis of Hospitals in 22 Countries

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Caesarean Sections in Teaching Hospitals: Systematic Review and Meta-analysis of Hospitals in 22 Countries

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Keywords

caesarean section, teaching hospitals, access to care, medical practice variation, health services, quality of care, overuse

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Abstract

Objective: The aim of this study is to determine the odds of caesarean section in all births in teaching hospitals as compared with non-teaching hospitals.

Setting: Over 3,600 teaching and non-teaching hospitals in 22 countries. We searched CINAHL, The Cochrane Library, PubMed, sciELO, Scopus and Web of Science from the beginning of records until May 2020.

Participants: Women at birth. Over 18.5 million births.

Intervention: Caesarean section.

Primary and Secondary Outcome Measures: The primary outcome measures are the adjusted odds ratio of caesarean section in a variety of teaching hospital comparisons. The secondary outcome is the crude odds ratio of caesarean section in a variety of teaching hospital comparisons.

Results: In adjusted analyses, we found that university hospitals have lower odds than non-teaching hospitals (OR=0.66, 95%CI 0.56, 0.78) and other teaching hospitals (OR=0.46, 95%CI 0.24, 0.89), and no significant difference with unspecified teaching status hospitals (OR=0.92, 95%CI 0.80, 1.05, $\tau 2=0.009$). Other teaching hospitals had higher odds than non-teaching hospitals (OR=1.23, 95%CI 1.12, 1.35). Comparison between unspecified teaching hospitals and non-teaching hospitals (OR=0.91, 95%CI 0.50, 1.65, $\tau 2=1.007$) and unspecified hospitals (OR=0.95, 95%CI 0.76, 1.20), $\tau 2<0.001$) showed no significant difference. While the main analysis in larger sized groups of analysed studies reveals no effect between hospitals, subgroup analyses shows that teaching hospitals carry out fewer CS in several countries, for several study populations and population characteristics.

Conclusions: With smaller sample of participants and studies, in clearly defined hospitals categories under comparison, we see that university hospitals have lower odds for caesarean. With larger sample size and number of studies, as well as less clearly defined categories of hospitals, we see no significant difference in the likelihood of caesarean sections between teaching and non-teaching hospitals. Nevertheless, even in groups with no significant effect, teaching hospitals have a lower or higher

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likelihood of caesarean sections in several analysed subgroups. Therefore, we recommend a more precise examination of forces sustaining these trends.

Trial Registration: Prospero - CRD42020158437

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Strengths and limitations of this study

- The major strengths of our systematic review and meta-analysis include a broad literature search, data extraction performed by more than one reviewer, an exploration of study characteristics as a potential source of variation between studies, and quality assessment using QUIPS.
- Some additional strengths of this study are the large sample size and the inclusion of many hospitals over broad geographic regions.
- Potential limitations relate to differences in the characteristics of compared hospitals, study populations, type of data used, types of CS analysed, and variables used for adjustment in statistical analyses across studies.
- We also lack information on hospital resources, infrastructure and staffing of compared hospitals which could have been useful for a more detailed analysis of the underlying factors that may contribute to differences among teaching and non-teaching hospitals.

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Introduction

Caesarean sections (CS) are a key surgical procedure in improving long-term health outcomes in high-risk births.[1 2] Key to providing high quality healthcare is avoiding the risks associated with using this procedure in low-risk pregnancies. Despite recommendations by the WHO to decrease CS rates and perform CS based primarily on medical indications,[3-5] CS rates continue to increase, rising to an estimated 21.1% worldwide in 2015.[6] Furthermore, there is a large degree of variation observed. The rate in different countries ranges from dangerously low, i.e. below 10%, to extraordinary high rates, e.g. 58.1% in the Dominican Republic.[6] Within country variations have also been observed in many countries including the United States.[7] More recently, in Italy, the 2018 data concerning Nulliparous, Term, Singleton, Vertex CS shows variation between 8.4% and 44.4% among health authorities, with significant and constant intra-regional variation.[8]

It is highly unlikely that such dramatic differences are explained solely by clinical factors. They may stem from a variety of factors, such as insurance status or socioeconomic background.[9-13] Factors affecting patient or clinician preferences may also be responsible for variations in healthcare delivery. Patients prefer caesareans for cultural or experience related reasons. Physicians can be less skilled in relation to natural birth, in comparison to, for example, midwives.[14-21] Hospital factors or insurance type may also have an effect.[10 22 23]

Hospital teaching status is one of hospital characteristics that may explain the variability of CS among hospitals. In general, teaching hospitals are known to have higher overall rates of surgery, which is perhaps reflective of more complex caseloads.[24-26] For this reason, it may be intuitively assumed that teaching hospitals have a higher proportion of high-risk births and, as such, higher likelihood of CS. Teaching hospitals are generally associated with a higher standard of care and better outcomes.[26-32] There is more adherence to evidence-based guidelines and/or clinical protocols. Higher quality care, in turn, can also translate to a lower likelihood of caesarean delivery as a result of fewer unnecessary procedures. Teaching hospitals also generally have access to a certain level of technology, and it is easier to apply protective measures, such as requiring a second opinion on CS procedures.[27] The goal of this review is examine the odds of caesarean section in teaching and non-teaching hospitals. We aim to determine

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whether there is a significant difference, evaluate the effect size and variations among different groups and subgroups, and thereby understand the differences in delivery care provision between teaching and non-teaching hospitals.

Materials and Methods

This meta-analysis was constructed as per previous meta-analyses carried out by members of our group[22] [33 34 35] and in accordance with the structure suggested by Dekkers and co-workers,[36] the Preferred Reporting Items for Systematic reviews (PRISMA) guidelines,[37] and Meta-analysis of Observational Studies in Epidemiology (MOOSE) guidelines for meta-analysis of observational studies.[38] We produced a protocol, which was submitted and registered with Prospero (Prospero ID: CRD42020158437).

Eligibility Criteria, Information Sources, Search Strategy

In order to get the widest spread of data possible we searched CINAHL, The Cochrane Library, PubMed, sciELO, Scopus and Web of Science from the beginning of records until May 2020. The search terms were based on a four-level strategy, with terms related to caesarean sections, terms related to hospital teaching status, terms related to area of research i.e. health system variation etc., and terms related to study design.

Study Selection

The primary criterion for study inclusion was the presence of an odds ratio (OR) for caesarean sections in teaching and non-teaching hospitals. Reports which included other data (i.e. regression coefficients, case data by type of hospital and delivery outcome) allowing the ORs to be calculated were also included. Study inclusion was also contingent on study design, targeting in this case reviews, systematic reviews, cohort studies and cross-sectional studies. Studies were excluded to remove overlapping data. This refers to the situation where more than one identified article (during screening/full text review) seems to have used same data sets. Studies were also excluded in case they were not reported in English language. The eligibility of each study was assessed independently by at least two researchers, and disagreements were resolved by consultation.

Data Extraction

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Data extraction was carried out by at least two independent researchers for each study. Another reviewer performed a final review of all extracted data. Disagreements were resolved by discussion. A data extraction spreadsheet was constructed to facilitate the process, and included a number of pertinent data, such as publication data, effect estimate, confidence limits, main determinant being studied and sample size, among other data. For the purpose of classification of population of each study we extracted exclusion/inclusion criteria data that were used by authors in order to specify their study population. Such data enabled us to classify population for each study based on Robson system. Robson's ten group classification system is based on five main population features including obstetric history, onset of labour, presentation of features, number of neonates, and gestational age.[39] This is a standardized comparison method that is helpful and widely used to analyse trends and determinants of CS.[40]

Assessment of Risk of Bias

The risk of bias for included studies was assessed in terms of six domains – study participation, study attrition, prognostic factor measurement, study confounding, and statistical analysis and reporting, using the Quality in Prognosis Studies (QUIPS) tool.[41] Although it is a standardised tool for prognostic studies, it fits very well with overall structure of included papers and it has already been used it in methodologically similar reviews.[32,33 35]

Data Synthesis

The ORs were combined using standard inverse-variance random-effects meta-analysis, where an OR lower than one shows a lower likelihood of CS in teaching hospitals than non-teaching hospitals.[42]

Using definitions and classifications provided by Ayanian and Weissman[43] as well as definitions used by papers included in the review we specified three main categories of hospitals: University hospital, other teaching hospital and non-teaching hospital. University hospital is a major teaching hospital affiliated with a medical school.[43] It is the institution where residency programs take place, high-end care is provided and research activity takes place. Other teaching hospitals don't meet such criteria, but they do provide residency training.[43] Non-teaching hospitals are hospitals that do not provide venue for training of residents. In case studies were not clear on the definition of comparing hospitals and reference group we

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marked them and unspecified teaching or as unspecified hospital. In case where studies didn't specify to which type of teaching hospitals they were referring to we marked them as unspecified teaching hospital. In such cases it is highly likely, i.e. studies for US, that such category of hospitals includes both university and other teaching hospitals. In case where studies did not specify reference hospital, we marked them as unspecified hospital. Based on our assessment such categorization of hospitals is likely to include both non-teaching hospitals or other teaching hospitals or one of them.

As a result, based on classification of teaching hospitals and reference hospitals studies used, we analysed adjusted and crude estimates reported by studies independently in six different groups: university vs non-teaching hospital, university vs other teaching hospital, university vs unspecified hospital, other teaching vs non-teaching hospital, unspecified teaching hospital vs non-teaching hospital, unspecified teaching hospital vs unspecified hospital.

Subgroup analysis for adjusted estimates by country, study design, period of data collection, population by Robson groups and criteria, type of CS analysed, type of data used and level of (QUIPS) risk of bias domains was used to examine between-study heterogeneity and χ^2 test used to calculate p values for interaction among subgroups.

Statistical analysis was carried out using STATA, release V.15 (StataCorp).

Patient Involvement

No patients were involved in this study. We used data from published papers only.

Results

Study Selection

We identified a total of 5233 records: 344 from CINAHL, 1049 from the Cochrane Library, 1522 from PubMed; 651 from sciELO, 534 from Scopus; 1133 from Web of Science. Additional 68 recods were identified from a manual search (Figure 1). We removed 783 duplicates. 4518 records were screened for eligibility. We performed full-text examination on 908 records. We excluded 827 that did not report hospital teaching status, 46 that were otherwise irrelevant and three records with overlapping populations.

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Finally, 32 records describing 32 separate studies,[18 27 44-73] including 18.5 million births in 22 different countries, were included in the review and meta-analysis. Of these, 18 studies were included in the meta-analyses of adjusted odds ratios and 27 studies were used in the meta-analyses of crude odds ratios.

Study Characteristics

Characteristics of included studies are shown in Table 1 and Online Appendix. Most included studies were retrospective studies, with some prospective studies also included. The studies were from a variety of countries and health systems. Sample size varied from 214 to 8.48 million births. The oldest study was from 1991, but the vast majority of included studies were carried out post-2000. Case exclusion criteria and covariates for statistical adjustment also varied considerably.(Online Appendix)

Table 1 - Characteristics of Included Studies

Risk of Bias of Included Studies

The main potential source of bias is study confounding, where 17 of the 32 studies included had a high risk of bias and 9 studies had a moderate risk of bias. With respect to study participation, 4 studies had a moderate risk of bias, with the remainder having a low risk of bias. A low risk of bias in all studies was reported for study attrition, prognostic factor measurement and outcome measurement, while only two studies had a moderate risk of bias for the statistical analysis and reporting domain.

Synthesis of Results

Figure 2 presents meta-analyses for primary outcome measures, i.e. adjusted ORs for all six groups we analysed. In the first group, including one study, university hospitals had lower odds of CS than non-teaching hospitals (OR=0.66. 95%CI 0.56, 0.78). Similarly, in second group, university hospitals CS was 0.46 times as likely than in other teaching hospitals (95% CI 0.24 to 0.89). In the third comparison group involving three studies, university hospitals, did not have significantly different odds of CS compared to unspecified hospitals (OR=0.92 95% 0.80, 1.05) with low heterogeneity between studies ($\tau 2=0.009$). Analysis of the group of other teaching hospitals in comparison with non-teaching hospitals from one study showed higher odds of CS (OR=1.23, 95%CI 1.12,1.35). Comparison group of unspecified teaching hospitals with non-teaching hospitals, which incorporated 11 studies and was the largest adjusted

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analysis, showed no significant difference (OR=0.91, 95%CI 0.50, 1.65) with high heterogeneity between studies ($\tau 2=1.007$). Similarly, including three studies, unspecified teaching hospitals did not have significantly lower odds than unspecified hospitals (OR=0.95, 95%CI 0.76, 1.20) with no heterogeneity between studies ($\tau 2<0.001$).

Subgroup analyses of adjusted odds ratios in teaching and non-teaching hospitals (Online Appendix) showed some variation. For the comparison of university hospitals and unspecified hospitals, lower odds of CS were observed for France (OR=0.84, 95%CI 0.74, 0.97), while for USA the odds were the same (OR=1.00, 95%CI 0.95, 1.06). Lower odds were observed also in the subgroup with single number of fetuses, studies with low risk of confounding (as defined by QUIPS), studies using hospital data, as well as studies not using birth registry data.

Subgroup analysis for the comparison of unspecified teaching hospitals with non-teaching hospitals highlighted difference in many factors. Inter country variation was particularly notable, where France (OR=0.30, 95%CI 0.16, 0.58), Palestine (OR=0.53, 95%CI 0.37, 0.77), Taiwan (OR=0.86, 95%CI 0.82, 0.91) and US (OR=0.80, 95%CI 0.72, 0.88) had lower odds of CS. Conversely, Korea (OR=4.95, 95%CI 4.82, 5.09) and Lebanon (OR=1.40, 95%CI 1.30, 1.50) had higher odds of CS in teaching hospitals. There was also variation in Robson subgroups. Studies using groups 1 and 3 showed lower odds of CS in teaching hospitals (OR=0.80, 95%CI 0.69, 0.93), as did studies with mothers from groups 6 and 7 (OR=0.30, 95%CI 0.16, 0.58). Conversely, studies using all Robson subgroups had higher odds of CS (OR=4.95, 95%CI 4.82, 5.09). There was also variation in odds ratio with studies including different fetal presentation, where studies with any fetal presentation showed no difference (OR=1.04, 95%CI 0.50, 2.14), studies including breach presentation showed lower odds of CS (OR=0.30, 95%CI 0.16, 0.58) and odds of 0.88 were observed for cephalic presentation (95%CI, 0.75, 1.04). Studies with high (1.57, 95%CI 0.39, 6.32) and low (1.15, 95%CI 0.79, 1.69) QUIPS confounding showed no significant difference in odds, while studies with moderate risk of confounding had lower odds of CS (0.73, 95%CI 0.69, 0.78).

In secondary outcome analysis (Figure 3), CS in university hospitals was 1.36 times as likely as compared to non-teaching hospitals, although the difference was not significant (95%CI 0.80, 2.32, $\tau 2=0.208$), and

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0.50 times as likely in comparison with other teaching hospitals (95%CI 0.35, 0.71). In comparison with unspecified hospitals, there was again no significant difference (OR=1.01, 95%CI 0.86, 1.17) with low heterogeneity between studies (τ2=0.019). CS was 1.28 times as likely in other teaching hospitals compared to non-teaching hospitals (95%CI 1.11, 1.46) with no relevant heterogeneity between studies (τ2=0.009). Our unadjusted comparisons of unspecified types of teaching hospital with non-teaching hospitals and unspecified hospitals reflected the results of the adjusted analysis, with no significant difference observed. In the case of non-teaching hospitals CS was 1.01 times as likely (95%CI 0.63, 1.61) with high heterogeneity between studies (τ2=0.859). For unspecified teaching hospitals, in comparison with unspecified hospitals, there was no significant difference in the odds of CS (OR=0.96, 95%CI 0.85, 1.09) with no relevant heterogeneity between studies (τ2=0.006).

Discussion

In more clearly defined hospital comparison groups, we see lowering effect of teaching hospitals as compared with non-teaching hospitals. But samples and number of studies are small. In most of the studies and largest sample size we see no overall significant difference between teaching and non-teaching hospitals in adjusted and subgroup analyses. Subgroup analyses showed varied estimates.

Strengths and weaknesses

The major strengths of our systematic review and meta-analysis include a broad literature search, data extraction performed by more than one reviewer, an exploration of study characteristics as a potential source of variation between studies, and quality assessment using QUIPS. Some additional strengths of this study are the very large sample size and the inclusion of many hospitals over broad geographic regions. Potential limitations which may have affected inconsistencies in the results relate to differences in the characteristics of compared hospitals, study populations, type of data used, types of CS analysed, and variables used for adjustment in statistical analyses across studies. The fact that most studies have not been as precise in definition of specific categories of teaching status of hospital is particularly limiting. Moreover, we lack information on hospital resources, infrastructure and staffing, all of which could contribute to maternal outcomes and risk for CS.[7]

Context

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The findings in this study demonstrate the complexity of the interactions of many factors. The role of health system factors in general, and hospital factors in particular should not be ignored. Recent meta-analyses examining the effect of hospitals and financial factors[22 33 34] have found that CS was 1.41 times as likely in for-profit hospitals as compared with non-profit hospitals (95% CI 1.24 to 1.60),[22] 1.13 for privately insured women compared with women covered with public insurance (95% CI 1.07 to 1.18) [34] and 0.70 in uninsured as compared to privately and publicly insured women (95%CI 0.69, 0.72).[33]

Interpretation

The findings in our group and subgroup analyses indicate that the effect of hospital teaching status varies between countries, study design and confounders considered in data analysis. Specific populations of mothers, and particular type of CS are also important variables in examining the effect of hospital variation. Patient populations in teaching versus non-teaching hospitals are generally very different in terms of underlying medical conditions and obstetric risk, which would likely drive any differences observed in odds for CS. Hence, intuitively, we would expect that odds for CS are likely to be higher in teaching hospitals, where more complex cases are often taken. Our study does not show this, and in several groups and subgroups the opposite is observed. Variation in odds of CS in teaching hospitals for particular Robson groups hints at differences in implementation of clinical standards. Studies have suggested that the lower odds in teaching hospitals may be due to availability of resources,[44 74] access to technology,[75] closer supervision and care,[60] stronger accountability[76], adherence to clinical standards,[18 27 57 77 78] and professional consultation opportunities.[27]

Implications

Our study contributes to understanding of the role of hospital related causes on the rates of CS. It may also provide an impetus for further research, particularly setting-specific studies, or even individual patient data meta-analysis. Such studies would allow a more precise and in-depth analysis of the characteristics of hospitals associated with teaching status, such as resource availability or adherence to clinical guidelines. In order to improve the quality of healthcare provision for deliveries, the interplay of factors related to teaching status of hospitals must be examined, in all outcomes of related to deliveries, including but not limited to overall caesarean section, emergency and primary caesarean, vaginal birth

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after caesarean. Using classification by Robson groups or primary CS will help in understanding better the implications of results in future studies. Using variables with information such as infrastructure and other facets of hospital variation could be useful to elucidate more clearly the mechanism. Another way to understand the underlying factors related to hospital characteristics, which is being applied in Tuscany (Italy), is to put in place systematic and ongoing collection of PREMs (Patient-Reported Experience Measures) and PROMs (Patient-Reported Outcome Measures) [79] which could ensure continuous source of data to measure hospital performance outcomes.

Conclusion

With smaller sample of participants and studies, in clearly defined hospitals categories under comparison, we see that university hospitals have lower odds for caesarean. With larger sample size and number of studies, in less clearly defined categories of hospitals, we see no significant difference in the likelihood of caesarean sections between teaching and non-teaching hospitals. Nevertheless, even in groups with no significant effect, teaching hospitals in many subgroups, have a lower likelihood of caesarean sections. Lower or higher odds for CS among teaching hospitals hint at the effect of important aspects of healthcare delivery. Hence, further research would help to understand the interplay of underlying factors in individual settings which may provide useful insights for policy or medical practice efforts to ensure appropriate and efficient use of delivery care.

Author contributions

IH conceived and designed the study. BK, JB, FS performed the screening of studies. IH, EZ, KG, BK, JB, FS performed the data extraction and preparation. IH and EZ performed quality assessment of studies. IH has analysed the data. DAL, MB and GL have contributed to the interpretation of findings. IH, EZ and RA drafted the paper, which was critically reviewed and approved by all authors.

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All authors have completed the ICMJE uniform disclosure form at www.icmje.org/coi_disclosure.pdf and declare: no support from any organization for the submitted work; no financial relationships with any organizations that might have an interest in the submitted work in the previous three years; no other relationships or activities that could appear to have influenced the submitted work.

Data availability statement

All data relevant to the study are included in the article or uploaded as supplementary information.

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Table 1. Characteristics of included studies

Study	Year	Country	Study design	Number of cases	Number of hospital units	Year of data collection	Period of data collection	Population by Robson groups	Sampling	Type of CS analysed
University vs non-teaching hospital										
Murray[48]	2000	Chile	Prospective	504	3	1995-1997	Before 2000	All	Random	Any
Garcia et al.[50]	2001	United States	Retrospective	51,009	36	1996	Before 2000	All	Consecutive	Any
Zhang et al.[58]	2010	United States	Retrospective	132,667	19	2002-2008	2000 and after	Multiple	Consecutive	Any
University vs other teaching hospital										
Hoxha et al.[18]	2019	Kosovo	Prospective	859	5	2015	2000 and after	1 to 4	Consecutive	Any
University vs unspecified hospital										
David et al.[49]	2001	France	Retrospective	40,512	149	1994-1998	Before 2000	Multiple	Consecutive	Any
Carayol et al.[53]	2007	France and Belgium	Prospective	6,091	174	2001-2002	2000 and after	Multiple	Random	Planned
Snyder et al.[60]	2011	United States	Retrospective	244,464	129	2006-2007	2000 and after	Multiple	Consecutive	Any
Oner et al.[66]	2016	Tukey	Prospective	214	4	2012	2000 and after	Multiple	Random	Any
Faisal-Cury et al.[67]	2017	Brazil	Prospective	757	10	2005-2006	2000 and after	Multiple	Random	Any
Other teaching vs non-teaching hospital										
Garcia et al.[50]	2001	United States	Retrospective	58,771	36	1996	Before 2000	All	Consecutive	Any
Zhang et al.[58]	2010	United States	Retrospective	81,339	19	2002-2008	2000 and after	Multiple	Consecutive	Any

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Unspecified teaching hospital vs non-teaching hospital										
Sanchez-Ramos[44]	1994	United States	Prospective	945,707	236	1992	Before 2000	All	Random	Any
Braveman et al.[45]	1995	United States	Retrospective	217,461	Not reported	1990-1991	Before 2000	Multiple	Consecutive	Any
Irwin et al.[46]	1996	United States	Retrospective	3,603	75	1987-1989	Before 2000	Multiple	Consecutive	Any
Chanrachakul et al.[47]	2000	Thailand	Retrospective	1,073,403	227	1999	Before 2000	All	Random	Any
Korst et al.[51]	2005	United States	Retrospective	443,532	288	1995	Before 2000	Multiple	Consecutive	Any
Linton et al.[52]	2005	United States	Retrospective	53,215	66	2002	2000 and after	All	Consecutive	Any
Tamim[54]	2007	Lebanon	Retrospective	18,837	9	2001-2002	2000 and after	Multiple	Consecutive	Any
Xirasagar et al.[55]	2007	Taiwan	Retrospective	739,531	942	1997-2000	Before 2000	Multiple	Consecutive	Planned
Chen et al.[56]	2008	Taiwan	Retrospective	200,207	Not reported	2004	2000 and after	Multiple	Consecutive	Any
Liu et al.[57]	2008	Taiwan	Retrospective	247,751	Not reported	1998-2002	Before and after 2000	Multiple	Consecutive	Any
Taljaard et al.[59]	2009	South America	Prospective	95,164	119	2004-2005	2000 and after	Multiple	Random	Any
Aman et al.[61]	2014	Ethiopia	Retrospective	17,761	6	2011	2000 and after	All	Consecutive	Any
Stivanello et al.[62]	2014	Italy	Retrospective	213,539	36	2005-2010	2000 and after	All	Consecutive	Any
Bommarito et al.[63]	2016	United States	Retrospective	8,485,034	1,051	1998-2010	Before and after 2000	All	Random	Emergency
Kim et al.[64]	2016	Korea	Retrospective	1,289,989	674	2011-2014	2000 and after	All	Consecutive	Any

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Lorthe et al.[68]	2017	France	Prospective	528	546	2011	2000 and after	6 and 7	Consecutive	Any
Clapp et al.[69]	2018	United States	Retrospective	1,657,495	1,206	2013	2000 and after	1 to 4	Consecutive	Any
Zimmo et al. A[72]	2018	Palestine	Prospective	18,908	3	2016-2017	2000 and after	All	Consecutive	Any
Zimmo et al. B[71]	2018	Palestine	Prospective	9,567	2	2015-2016	2000 and after	Multiple	Consecutive	Emergency
del Carmen et al.[73]	2019	United States	Retrospective	1,855,675	Not reported	2006-2010	2000 and after	1 and 3	Consecutive	Emergency
Unspecified teaching hospital vs unspecified hospital										
Oleske et al.[27]	1991	United States	Retrospective	130,249	198	1986	Before 2000	Multiple	Consecutive	Planned
Meda et al.[65]	2016	Burkina Faso	Retrospective	668	66	2009-2010	2000 and after	All	Random	Any
Lorthe et al.[68]	2017	France	Prospective	347	546	2011	2000 and after	6 and 7	Consecutive	Any
Zbiri et al.[70]	2018	France	Retrospective	291,449	11	2008-2014	2000 and after	All	Consecutive	Any

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Figure legends

Figure 1. The flow diagram of review

Figure 2. Adjusted odds ratios of caesarean section

Figure 3. Crude odds ratios of caesarean section

Supporting information

Appendix with detailed study information and subgroup analysis

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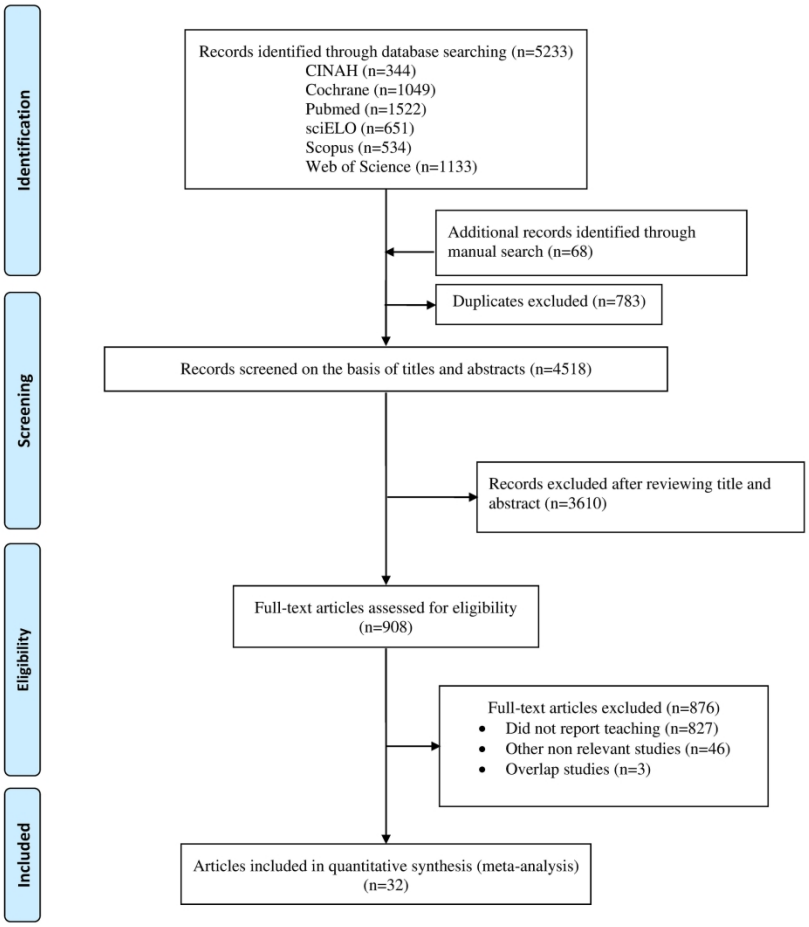


Figure 1. The flow diagram of review

215x279mm (300 x 300 DPI)

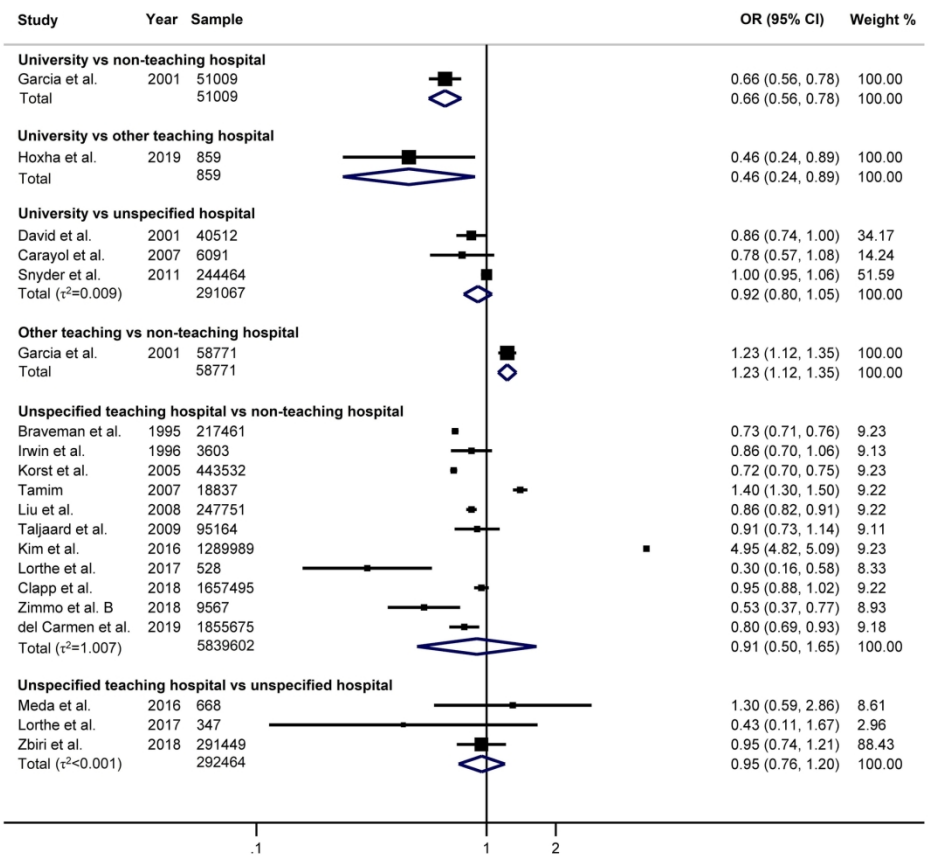


Figure 2. Adjusted odds ratios of caesarean section

172x166mm (300 x 300 DPI)

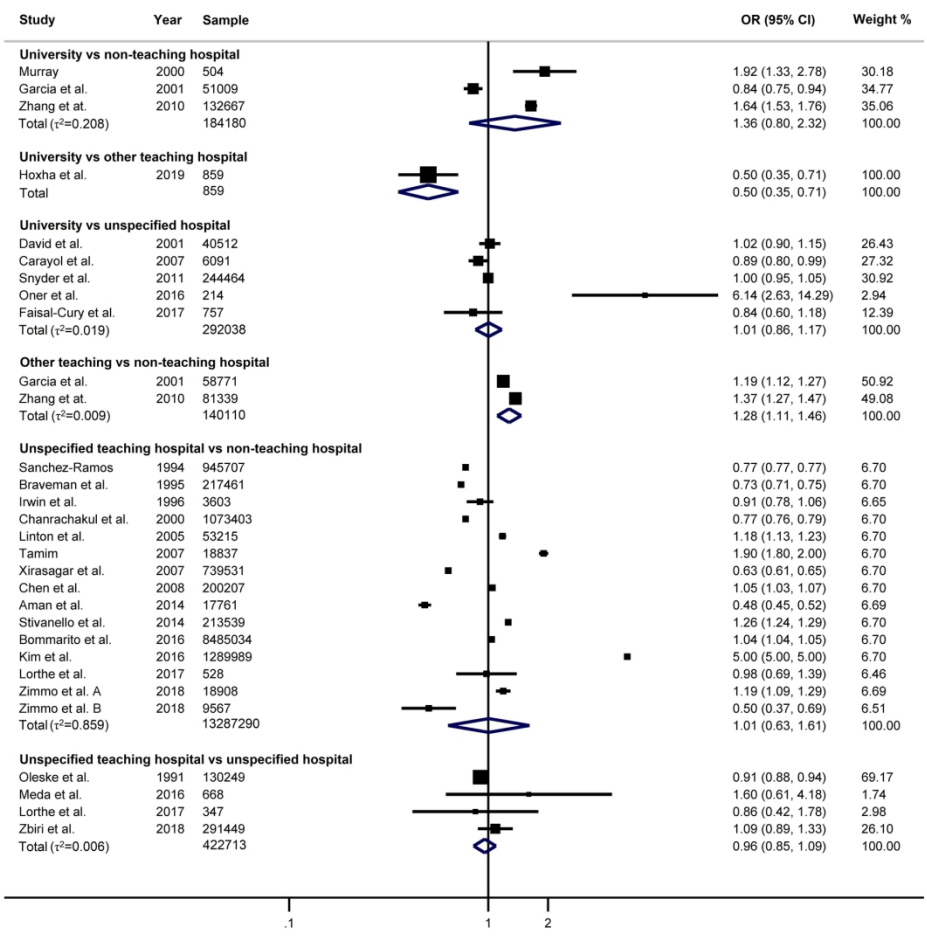


Figure 3. Crude odds ratios of caesarean section

174x181mm (300 x 300 DPI)

Study	Year	Type of data used: Survey	Hospital records	Birth certificates/registry	Claims data	Source population
University vs non-teaching hospital						
Murray[48]	2000	x				All births in 2 hopitals in Santiago, Chile
Garcia et al.[50]	2001		x		x	All births in 36 hospitals in Maryland, United States
Zhang et at.[58]	2010		x			All births in 12 clinical centres in United States
University vs other teaching hospital						
Hoxha et al.[18]	2019	x	x			All births in 5 public hospitals in Kosovo
University vs unspecified hospital						
David et al.[49]	2001		x			All births in 149 hospitals in France
Carayol et al.[53]	2007	x	x			All births in 138 hospitals in France and 36 hospitals in Belgium
Snyder et al.[60]	2011			x		All births in 129 hospitals in Ohio, United States
Oner et al.[66]	2016	x				All births in 82 hospitals in Burdur Province, Turkey
Faisal-Cury et al.[67]	2017	x	x			All births in 10 hospitals in Sao Paolo, Brazil
Other teaching vs non-teaching hospital						
Garcia et al.[50]	2001		x		x	All births in 36 hospitals in Maryland, United States
Zhang et at.[58]	2010		x			All births in 12 clinical centres in United States
Unspecified teaching hospital vs non-teaching hospital						
Sanchez-Ramos[44]	1994	x				All births is 236 hospitals in United States
Braveman et al.[45]	1995			x		All births in California, United States
Irwin et al.[46]	1996		x			All births in 75 military hospitals in United States
Chanrachakul et al.[47]	2000	x				All births in 227 hospitals inThailand
Korst et al.[51]	2005		x			All births in California, United States
Linton et al.[52]	2005		x			All births in 66 military hospitals in United States
Tamim[54]	2007	x	x			All births in 9 hospitals in Beirut, Lebanon
Xirasagar et al.[55]	2007				x	All births in 942 hospitals in Taiwan
Chen et al.[56]	2008				x	All births in Taiwan
Liu et al.[57]	2008					All births in Taiwan
Taljaard et al.[59]	2009	x				All births in 119 hospitals in South America
Aman et al.[61]	2014		x			All births in 6 hospitals in Addis Ababa, Ethiopia
Stivanello et al.[62]	2014		x	x		All births in 36 hospitals in Emilia-Romagna, Italy
Bommarito et al.[63]	2016		x			All births in 1051 hospitals in United States
Kim et al.[64]	2016					All births in 674 hospitals in Korea
Lorthe et al.[68]	2017		x			All births in 546 hospitals in France
Clapp et al.[69]	2018		x			All births in 1206 hospitals in 21 states in United States
Zimmo et al. A[72]	2018					All births in 3 hospitals in Gaza, Palestine
Zimmo et al. B[71]	2018					All births in 2 hospitals in Palestine
del Carmen et al.[73]	2019		x		x	All births in California, United States
Unspecified teaching hospital vs unspecified hospital						
Oleske et al.[27]	1991		x			All births in 198 hospitals in Illinois, United States
Meda et al.[65]	2016	x				All births in Burkina Faso
Lorthe et al.[68]	2017		x			All births in 546 hospitals in France
Zbiri et al.[70]	2018			x		All births in 11 hospitals in France

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	Population by Robson criteria						Population by Robson groups									
Study	Parity	Previous CS	Number of fetuses	Fetal presentation or lie	Gestational age	Onset of labour (induction)										
University vs non-teaching hospital																
Murray[48]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Garcia et al.[50]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Zhang et al.[58]	Any	No	Any	Any	Any	Any	1	2	3	4		6	7	8	9	10
University vs other teaching hospital																
Hoxha et al.[18]	0	No	1	Cephalic	>=37	Any	1	2								
University vs unspecified hospital																
David et al.[49]	0	Any	1	Any	Any	Any	1	2				6			9	10
Carayol et al.[53]	Any	No	1	Cephalic and breach	>=37	Any	1	2	3	4		6	7			
Snyder et al.[60]	Any	Any	Any	Any	>=37	Any	1	2	3	4	5	6	7	8	9	
Oner et al.[66]	0	No	Any	Any	Any	Any	1	2				6		8	9	10
Faisal-Cury et al.[67]	Any	Any	1	Any	Any	Any	1	2	3	4	5	6	7		9	10
Other teaching vs non-teaching hospital																
Garcia et al.[50]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Zhang et al.[58]	Any	No	Any	Any	Any	Any	1	2	3	4		6	7	8	9	10
Unspecified teaching hospital vs non-teaching hospital																
Sanchez-Ramos[44]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Braveman et al.[45]	0	No	1	Any	Any	Any	1	2				6			9	10
Irwin et al.[46]	0	No	1	Any	Any	Any	1	2				6			9	10
Chanrachakul et al.[47]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Korst et al.[51]	Any	No	Any	Any	Any	Any	1	2	3	4		6	7	8	9	10
Linton et al.[52]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Tamim[54]	0	Any	Any	Any	Any	Any	1	2				6		8	9	10
Xirasagar et al.[55]	Any	Any	1	Any	>=37	Any	1	2	3	4	5	6	7		9	
Chen et al.[56]	Any	Any	1	Any	Any	Any	1	2	3	4	5	6	7		9	10
Liu et al.[57]	Any	Any	1	Any	Any	Any	1	2	3	4	5	6	7		9	10
Taljaard et al.[59]	Any	Any	Any	Any	>=37	Any	1	2	3	4	5	6	7	8	9	
Aman et al.[61]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Stivanello et al.[62]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Bommarito et al.[63]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Kim et al.[64]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Lorthe et al.[68]	Any	Any	1	Breach	Any	Any						6	7			
Clapp et al.[69]	Any	No	1	Cephalic	>=37	Any	1	2	3	4						
Zimmo et al. A[72]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Zimmo et al. B[71]	Any	Any	1	Any	Any	Any	1	2	3	4	5	6	7		9	10
del Carmen et al.[73]	Any	No	1	Cephalic	>=37	Spontaneo	1		3							
Unspecified teaching hospital vs unspecified hospital																
Oleske et al.[27]	Any	No	Any	Any	Any	Any	1	2	3	4		6	7	8	9	10
Meda et al.[65]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10
Lorthe et al.[68]	Any	Any	1	Breach	Any	Any						6	7			
Zbiri et al.[70]	Any	Any	Any	Any	Any	Any	1	2	3	4	5	6	7	8	9	10

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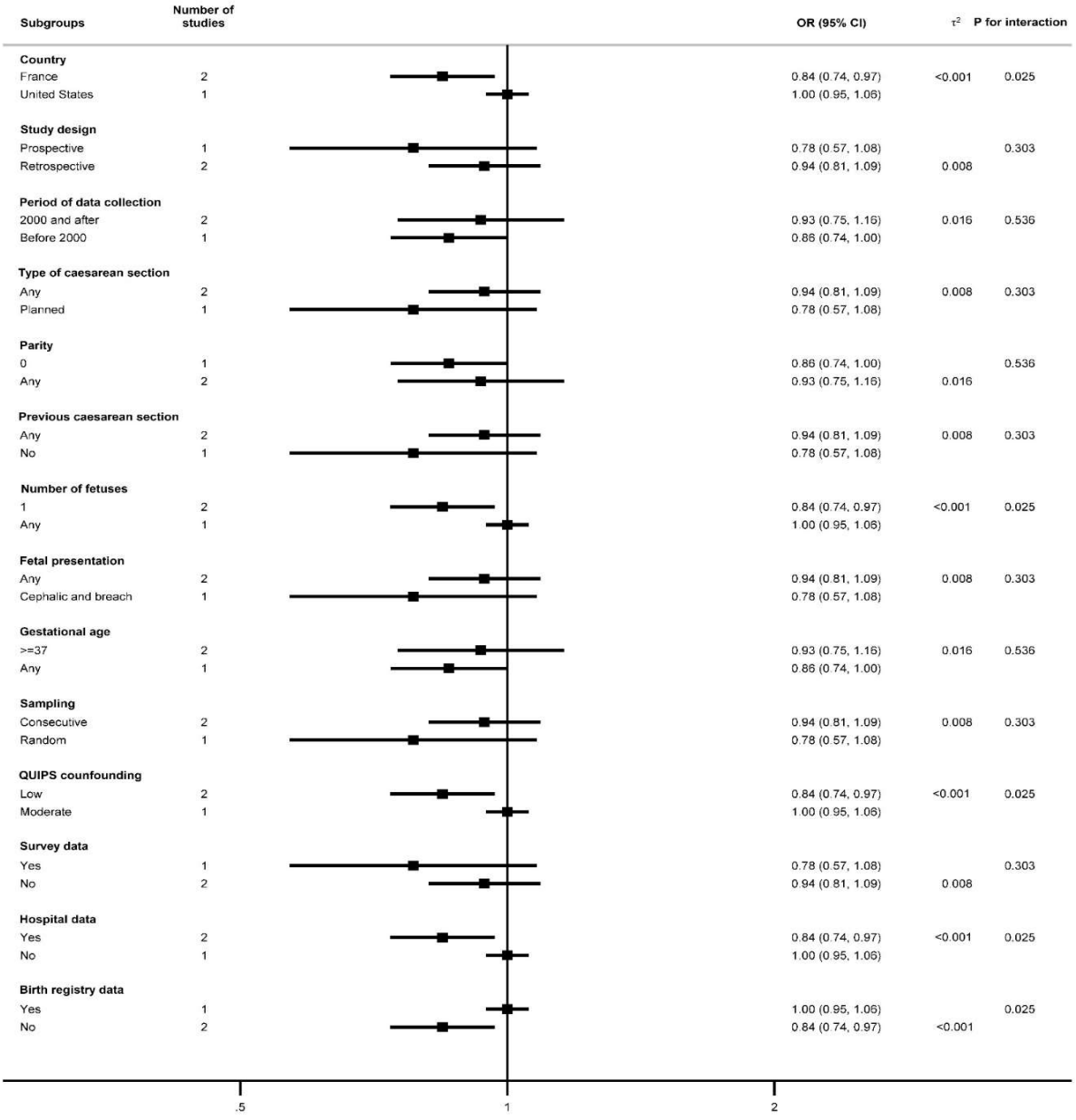
	Covariates used for statistical adjustment																						QUIPS risk of bias					
	Maternal preconception status											Maternal clinical status				Foetus characteristics												
Study	Ethnicity/Race	Educational level	Marital status	Economic status	Insurances status	Urban status	Weight	Height	Body mass index	Age	Parity	Previous caesarean section	Pre-existing (before pregnancy) conditi	Complications developed during pregna	Gestational age	Birth weight	Other characteristics	Prenatal care	Birth characteristics	Provider characteristics	Other variables	Total number of covariates	Study Participa	Study Attrition	Prognostic Fact	Outcome Meas	Study Confoun	Statistical Anal
University vs non-teaching hospital																												
Murray[48]																						0	moderate	low	low	low	high	low
Garcia et al.[50]	x		x		x	x				x		x								xx	xx	11	low	low	low	low	high	low
Zhang et al.[58]																						0	low	low	low	low	high	low
University vs other teaching hospital																												
Hoxha et al.[18]		x		x	x	x				x						x		x		xx	xx	12	moderate	low	low	low	low	low
University vs unspecified hospital																												
David et al.[49]										x	x	x	x	x	x				xx	xx	xx	16	low	low	low	low	low	low
Caravol et al.[53]		x								x	x										x	4	low	low	low	low	low	low
Snyder et al.[60]												x	xx	xx			x					6	low	low	low	low	moderate	low
Oner et al.[56]																					xx	3	low	low	low	low	high	low
Faisal-Cury et al.[67]	x	x	x	x					x	x	x			x						x	xx	13	moderate	low	low	low	high	low
Other teaching vs non-teaching hospital																												
Garcia et al.[50]	x		x		x	x				x		x								xx	xx	11	low	low	low	low	high	low
Zhang et al.[58]																						0	low	low	low	low	high	low
Unspecified teaching hospital vs non-teaching hospital																												
Sanchez-Ramos[44]																						0	low	low	low	low	high	moderate
Braveman et al.[45]	x	x	x	x	x					x			x			x	xx	x		xx	x	15	low	low	low	low	moderate	low
Irwin et al.[46]	x	x	x											x						x	x	6	low	low	low	low	moderate	low
Chanrachakul et al.[47]																						0	low	low	low	low	high	moderate
Korst et al.[51]	x				x					x										xx		7	low	low	low	low	moderate	low
Linton et al.[52]												x	xx	xx			x				xx	12	low	low	low	low	low	low
Tamim[54]					x				x	x	x			x	x	x		x			xx	11	low	low	low	low	low	low
Xirasagar et al.[55]																						0	low	low	low	low	high	low
Chen et al.[56]				x		x				x		x								xx	xx	9	low	low	low	low	high	low
Liu et al.[57]										x										xx		5	low	low	low	low	high	low
Taljaard et al.[59]											x	x	x	x			xx			xx	xx	10	low	low	low	low	high	low
Aman et al.[61]																						0	low	low	low	low	high	low
Stivanello et al.[62]																						0	low	low	low	low	high	low
Bommarito et al.[63]					x	x				x		x	xx	xx			xx	x	xx	xx	xx	0	low	low	low	low	high	low
Kim et al.[64]																						6	low	low	low	low	high	low
Lorthe et al.[68]			x							x	x				x					xx	xx	15	low	low	low	low	moderate	low
Clapp et al.[69]				x	x					x			xx	xx						xx		14	low	low	low	low	low	low
Zimmo et al. A[72]																						0	low	low	low	low	high	low
Zimmo et al. B[71]		x				x			x	x		x	x					x	x		xx	10	low	low	low	low	moderate	low
del Carmen et al.[73]	x				x	x				x			xx	xx			x		x	xx	xx	17	low	low	low	low	moderate	low
Unspecified teaching hospital vs unspecified hospital																												
Oleske et al.[27]					x					x										x	xx	5	low	low	low	low	moderate	low
Meda et al.[65]						x				x	x									xx	xx	7	moderate	low	low	low	high	low
Lorthe et al.[68]			x							x	x				x					xx	xx	15	low	low	low	low	moderate	low
Zbiri et al.[70]										x	x	x			x	x	x		x	xx	x	17	low	low	low	low	moderate	low

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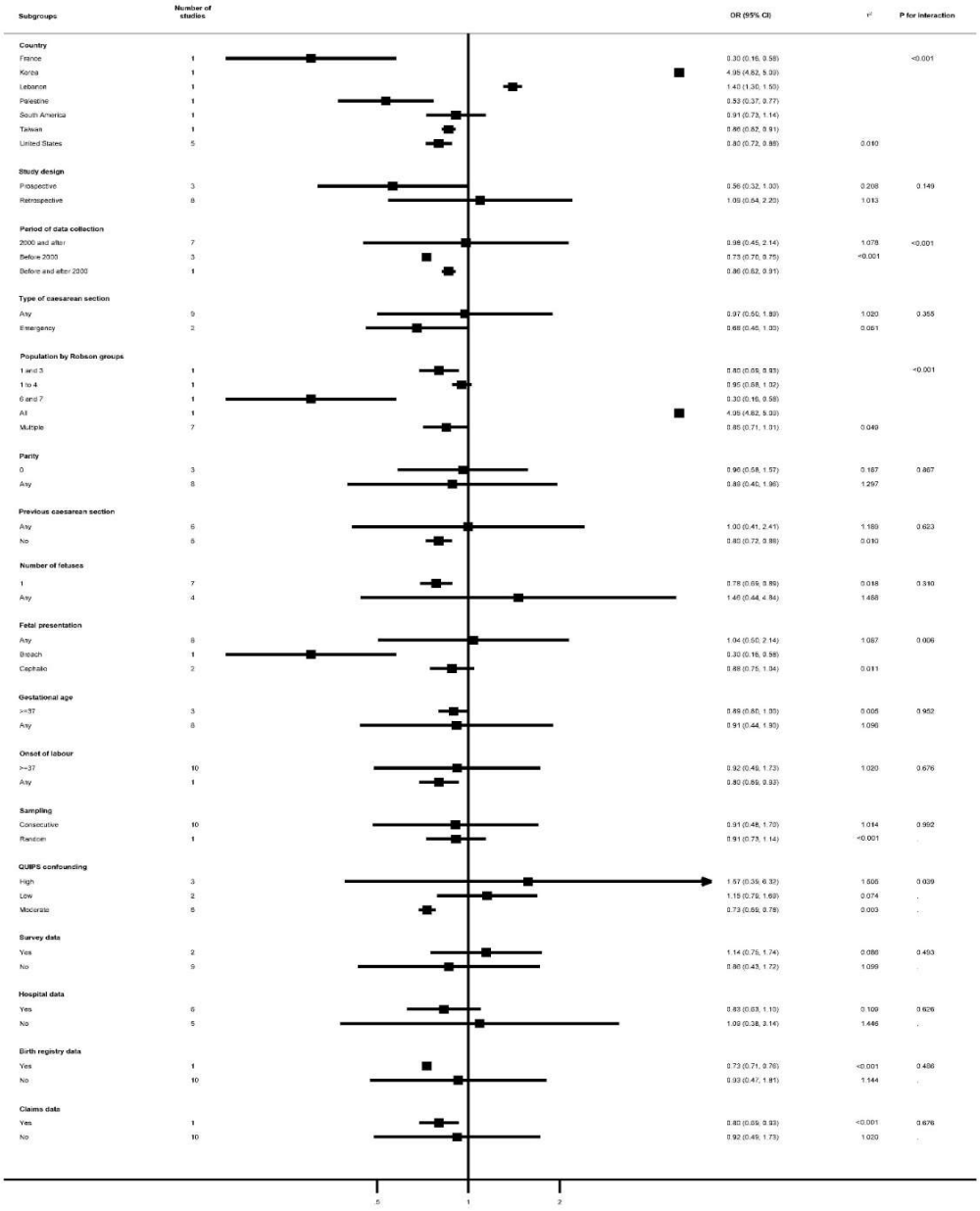
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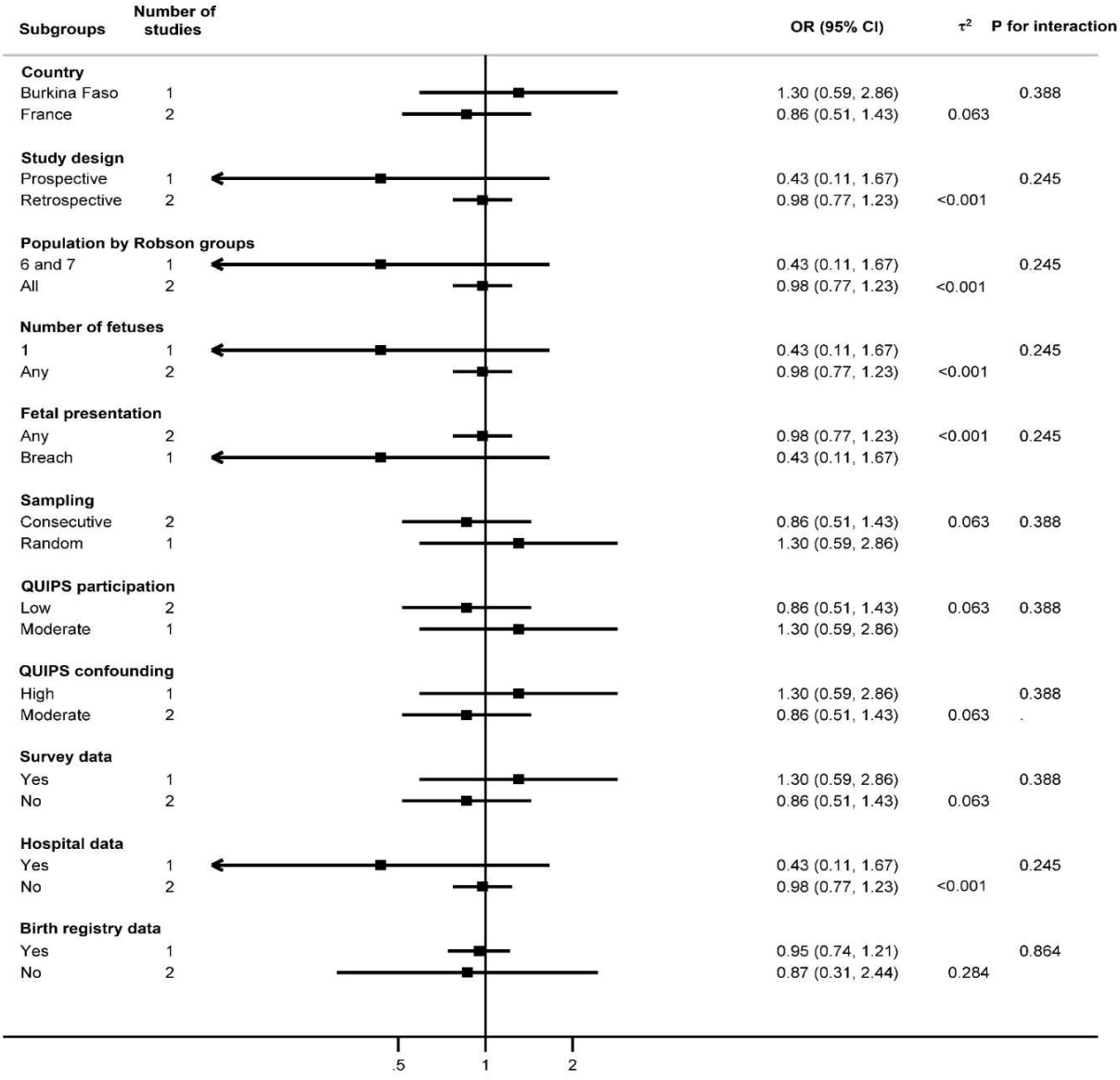
Subgroup analysis for university vs unspecified hospital group



Subgroup analysis for unspecified teaching hospital vs non-teaching hospital group



Subgroup analysis for unspecified teaching hospital vs unspecified hospital group



PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
TITLE			
Title	1	Identify the report as a systematic review, meta-analysis, or both.	1
ABSTRACT			
Structured summary	2	Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number.	3
INTRODUCTION			
Rationale	3	Describe the rationale for the review in the context of what is already known.	4
Objectives	4	Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS).	4,5
METHODS			
Protocol and registration	5	Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number.	5
Eligibility criteria	6	Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale.	5
Information sources	7	Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched.	5
Search	8	Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated.	5
Study selection	9	State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis).	5
Data collection process	10	Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators.	6
Data items	11	List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made.	6
Risk of bias in individual studies	12	Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis.	6
Summary measures	13	State the principal summary measures (e.g., risk ratio, difference in means).	6
Synthesis of results	14	Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., I^2) for each meta-analysis.	6

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PRISMA 2009 Checklist

Section/topic	#	Checklist item	Reported on page #
Risk of bias across studies	15	Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies).	6
Additional analyses	16	Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified.	6,7
RESULTS			
Study selection	17	Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram.	7
Study characteristics	18	For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations.	7
Risk of bias within studies	19	Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12).	7,8
Results of individual studies	20	For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot.	8
Synthesis of results	21	Present results of each meta-analysis done, including confidence intervals and measures of consistency.	8,9
Risk of bias across studies	22	Present results of any assessment of risk of bias across studies (see Item 15).	(Appendix)
Additional analysis	23	Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]).	8,9
DISCUSSION			
Summary of evidence	24	Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers).	9
Limitations	25	Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias).	9
Conclusions	26	Provide a general interpretation of the results in the context of other evidence, and implications for future research.	9-11
FUNDING			
Funding	27	Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review.	11-12

From: Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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